

Consider a partially ordered set P .

Call a nonempty partially ordered set $D \subseteq P$ *countably directed* if every countable subset of D has an upper bound in D .

Call $k \in P$ **$\aleph_1$ -compact** if for every countably directed $D \subseteq P$

$$k \leq \bigvee D \implies k \leq d \text{ for some } d \in D.$$

Consider a construction from descriptive set theory, namely the Baire space of weight $\aleph_1$, $(\omega_1)^\mathbb{N}$, where each copy of the first uncountable ordinal ω_1 has the discrete topology, and $(\omega_1)^\mathbb{N}$ has the product topology.

Consider the complete lattice L of all open sets of $(\omega_1)^\mathbb{N}$

Theorem. All elements of L except for the empty set are not $\aleph_1$ -compact.

Proof.

The proof will rely on *countable choice* (a weak version of the axiom of choice). However, when we run Lean verification of this fragment, we'll allow to just use the axiom of choice, which might make things easier to verify.

Consider a non-empty open set $U \in L$. We are going to build an uncountable increasing chain of open sets, this chain will be countably directed, the union of sets in this chain will be equal to U , but all elements of this chain will be strict subsets of U (that is, strictly less than U in the complete lattice L). That will establish that U is not $\aleph_1$ -compact.

One of the bases of this topology is formed by cylinder sets. Think about elements of $(\omega_1)^\mathbb{N}$ as functions mapping $\mathbb{N}$ to (ω_1) . Each cylinder set is defined by finite $I \subseteq \mathbb{N}$ and by fixed values f_i for each $i \in I$, and consist of all functions $f: \mathbb{N} \rightarrow \omega_1$, such that for all $i \in I$, $f(i) = f_i$.

So, for indices outside of I , functions in a cylinder set in question take all possible values in ω_1 .

Since cylinder sets form a basis of topology, take some cylinder set $V \subseteq U$ defined by $I \subseteq \mathbb{N}$ and fixed values at those indices. Consider an index $j \notin I$. Functions from V take all possible values $f(j)$ in ω_1 .

Split the cylinder set V into uncountable set of non-intersecting cylinders V_ξ , with each V_ξ defined by the index set $I \cup \{j\}$ and $f_j = f(j) = \xi$.

Consider the following uncountable increasing chain. For each $\alpha < \omega_1$ define

$$U_\alpha = (U \setminus V) \cup \bigcup_{\xi < \alpha} V_\xi.$$

Cylinder sets are not only open, but also closed. To see this, observe that the complement of V is a union of cylinder sets with the same index set I and with at least one f_i different from that of V , therefore the complement of V is open. From that it is easy to see that each U_α is open.

Each U_α is strictly smaller than U , since it misses V_α .

Similarly, this increasing chain is monotonically strictly increasing: $\alpha < \beta$ implies U_α being a strict subset of U_β .

Countable choice implies that every countable collection of ordinals below ω_1 has an upper bound below ω_1 (and, in fact, implies the uncountability of ω_1). The technical name for this property is that ω_1 is a *regular ordinal*, and this implication is well known.

This, in turn, implies that a countable collection of various U_{α_k} has an upper bound U_α , where α is an upper bound of the countable collection α_k in ω_1 . Therefore, the collection of U_α .

So, starting with an arbitrary non-empty open set $U \in L$, we obtain an uncountable increasing chain of open sets U_α , this chain is countably directed, the union of sets in this chain is equal to U , but all elements of this chain are strict subsets of U (that is, strictly less than U in the complete lattice L). That establishes that U is not $\aleph_1$ -compact.

QED

Corollary. L is not $\aleph_1$ -algebraic.

The corollary is easy to establish. Since only the empty set is $\aleph_1$ -compact, it is the only element which can be obtained as exact upper bound of $\aleph_1$ -compact sets, all other elements of L cannot be obtained in this fashion.